

BANARAS HINDU UNIVERSITY

B.Sc. (Hons) Semester I (NEP) Examination 2024-25

SEC — CHEMISTRY**Paper No. CHESE11 : Quality Control in Chemical Analysis**

Time: 2:30 Hours

Maximum Marks: 70

(Write your Roll No. at the top immediately on the receipt of this question paper)

Note: Attempt five questions including Question 1 which is compulsory. The figures in the right-hand margin indicate marks.

1. Answer all of the following:

- (a) Define and differentiate between the following: [5 × 2 = 10]
- (i) Standard deviation and Relative standard deviation of a data set.
 - (ii) Procedure and Protocol.
 - (iii) Determinate and Indeterminate errors.
 - (iv) Limit of Detection (LOD) and Limit of Quantification (LOQ).
 - (v) ISO and ISI.
- (b) Define 'Quality' and 'Total Quality Management' (TQM). [2]
- (c) Comment on the economic benefits of using 'standards' in analytical laboratories. [2]

2. Answer the following:

- (a) Following are the results of iron content analysis of a mineral: 14.50, 14.55, 14.26, 14.56, 14.53, 14.54. Calculate the following parameters for the above analysis: [10]
- (i) Standard deviation
 - (ii) Relative standard deviation
 - (iii) Variance
 - (iv) Coefficient of variation
- (b) How can determinate errors be minimized in analytical techniques for accurate analysis? [4]

3. Answer the following:

- (a) State any two principles of quality management. [2]
- (b) What is the 'Six Sigma' approach to quality management? Discuss its five steps (DMAIC). [6]

- (c) Why is it necessary to implement a quality management system for a pharmaceutical manufacturing unit? Explain. [6]

4. Answer the following:

- (a) What does 'Kaizen' mean in the context of quality management? State the basic principle on which it works. [4]
- (b) Why are ISO 9000 and ISO 14000 families known as 'generic management system standards'? Highlight the key differences between ISO 9000 and ISO 14000. [6]
- (c) What is the criterion for rejection of an observation/datum in a set of analytical data? [4]

5. Answer the following:

- (a) Explain the analytical validation of results for any analytical estimation method. [6]
- (b) What is a normal error curve? Draw and give its mathematical expression, significance, and show the standard deviation (σ) limits in the curve. [8]

6. Answer the following:

- (a) Explain ICH guidelines for the approval of any pharmaceutical drug for human use. [5]
- (b) Elaborate on the role and responsibilities of 'Quality Units' in a pharmaceutical manufacturing unit as per ICH guidelines. [4]
- (c) Give the key features of 'Good Manufacturing Practices' (GMP) adapted for the production of a pharmaceutical drug. [5]

7. Answer the following:

- (a) What is 'sampling'? [3]
- (b) How will you classify samples: (i) as per size, and (ii) as per constitution of analyte? [5]
- (c) How does a 'laboratory' sample differ from a 'real' sample? [3]
- (d) Write the various ways of expressing 'accuracy' and 'precision' of a set of analytical data. [3]